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Design of a short-range remote methane detector

By: Cheung Tsz Ho

Supervisor: Prof Ren Wei

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Abstract

1. This report presents the design of a remote methane detector/ TDLAS experiment using the principles and knowledge of optical spectroscopy and gas diagnostics as detailed in the textbook “Spectroscopy and Optical Diagnostics for Gases” by Ronald K. Hanson, R. Mitchell Spearrin, and Christopher S. Goldenstein [1].
2. This project aims to design the remote methane detector for bringing a convenient solution to detect the hidden gas in the industrial environment and monitoring the industrial safety.
3. The testing and lab experiment will be using the gas methane (CH_4), which is a greenhouse gas and key component some of the industrial environment, for example, construction site for building, gas station, etc.
4. To more close to the real industrial situation, the results and the discussions will based the on the experiment, which is testing the methane at normal pressure and ambient situation in the gas cell, the more professional vacuum gas cell testing and discussions will be made if having the chance to improve and enhance the accuracy in the future.

Acknowledgement

First, I would like to express my sincere thankful to my project supervisor, Prof. Ren Wei, for his invaluable guidance, support, and advice throughout this Final Year Project. His expertise in optical diagnostics and spectroscopy has been instrumental for me in shaping this work.

Second, I am also thankful to my tutor, XIAO XunZhou, for giving me the technical advice for shaping the system, setting up the devices of this Final Year Project.

Finally, I would like to acknowledge the use of the textbook “Spectroscopy and Optical Diagnostics for Gases” by Ronald K. Hanson, R. Mitchell Spearrin, and Christopher S. Goldenstein as the primary reference for the theoretical foundation of this TDLAS methane detection experiment.

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Problem description

1. Methane, which is an invisible, odourless and highly flammable gas, that works as a key component of the natural gas, as the natural gas works and exist in various site, for example, construction site, gas station, etc. Which will increase the risk of the people who works and living nearby, such, providing a solution for detecting the methane leaking is a serious and important thing for protecting and helping the society.
2. As compared to the traditional solution for detecting methane leaking, soap bubbling test, laser leak detector [2], the remote detector use in this project is more convenient and the function will be more perfect, as soap bubbling only can test with locating the connections and pipes, and laser leak detector only works for methane [2], shows that they will have many limitations that limit their usage and functions, but the remote methane detector for this project works if the air located at the suspicious leaking site can insert to the pipe to detect the leakage. also, while we change the IR laser and the wedge window to a suitable wavelength, we also can detect some of the other toxic gases if they are also leaking in there.

Result of literature review and Background Knowledge

1. The system mainly utilizing with the infrared laser source (IR), gas cell system, and also with the receiver, by the laser beam passes through the gas cell, the gas in the cell will absorb the IR laser, by the vibrational transitions in heteronuclear molecules, and while the vibration causing a change in the electric dipole moment [1], which will make the gas to absorb the laser, and while in this moment, we can use the receiver to represent the data with the intensity or power vs the time or the voltage of the receiver, to show whether the gas exist in the gas cell or not, at the same time, the absorbance can be calculated by the Beer-Lambert Law.

The design of the gas cell, the gas cell is choose to use the circular shape as the main body, at the two sides of the cell, can be used to install the mounting plate, there is a slot for each of the plate, which is use for putting the O-ring inside, and sealing and install the window, to prevent the leakage of the gas inside the cell.

While at the top of the cell there are two NPT type thread hole, which is for the male connector [3], for inserting and releases the gas in and out of the cell.

At the bottom there is one M4 size thread hole, which is use for holding the entire cell, to make it more stable while using the cell to do the experiment.

Also the grade of the screw using is 12.9, compare to the normal screws, the grade 12.9 thread will have a much more stable and highest strength grade compare to the other thread in the market using.

2. For the material use, first the material of the cell use will be Aluminium, there are several to choose to use Aluminium as the material of the gas cell.

First, aluminium can be a high-cost performance ratio of the solution to the gas cell, since the aluminium can be customized the design using 3D drawing software, for example fusion 360, SolidWorks, etc. And compare to other material to fabricate, stainless steel will be too expensive, and the PLA, ABS 3d printing material, the dimension will have a much more error compare to aluminium, also those traditional 3D printing material cannot ensure the air permeability, and the material status while in the vacuum environment, so,

choosing to use aluminium is a better choice compared to stainless steel and 3D printing material, with more suitability to this project use.

Second, aluminium has a good thermal conductivity, which is amount $237 \text{ Wm}^{-1}\text{K}^{-1}$, and low density compared to steel and cast iron [4], which means that the aluminium will have a good heat transfer rate, in some of the situation, for example, hot construction site, it can transfer the heat out to the surroundings quickly, and the temperature may increase the energy and intensity power of the gas, by formula $KE = 3/2RT$ shows, the temperature increase, the energy of gas will also increase, and it will affect the amount of energy that the molecules can absorb, since laser also is a type of energy sources, which will makes the molecules absorb less laser, and affect the accuracy of intensity power of laser that collect and represent using the receiver, such that using aluminium is a good choice for gas cell since have a great thermal conductivity

3. For the glass use in the gas cell, there will be using 2 high accuracy wedge windows, the high accuracy is needed for stabilize the laser input and output, to avoid any holes on the glass and the dimension is correct with less errors, also since we are choose to use the wedge glass, which the surface of the wedge glass are at a slight angle, to prevent the reflection of the laser back to the surroundings, as compare to the normal glass, the flat surface of the normal glass will reflect some of the laser back to the source site [5], such compare to wedge glass, using normal glass in this project may affect the accuracy of the data, such wedge glass is more preferred.
4. For the O-ring dimensions and the material use for the gas cell, due to squeezing of the force with the O-ring, the O-ring ID should 1-5% slight smaller than the groove [6], and the OD of the O-ring is better to $\geq$ the groove OD, such, by the formula, the O-ring ID is at most 12.125mm and the OD is more depends on the thickness of the O-ring. And by choosing the correct O-ring, from Taobao, which is 17OD x12.2ID and 2.4mm thickness, which can help to seal the glass and the tube to make it vacuum.

For the material use of O-ring, according to material chart selection [7], as we are assuming the gas tube will in vacuum situation, such Viton and Aflas type O-ring we are using.

For the height of the groove, since we are using axial face sealing, such, 20% or 30% squeeze is recommended [8], and by the formula of recommend groove height, $\text{squeeze\%} = (\text{Initial_Height} - \text{Squeeze_Height}) / \text{Initial_Height}$ [8], such that, after choosing the 12.2mm O-ring, the height is 2.4mm and the squeeze

percentage is 25%, then the groove height should be around $25\% = (2.4 - SH)/2.4$, and SH should be 1.8mm, which the height is decided for the O-ring and apply in the cad drawing.

The size of the wedge glass is increased from 12.5mm to 25mm due to the original height is too large for making the O-ring compress, and after the adjusting, the O-ring is squeezed and fit to the slot successfully, also, the surface will have a tiny angle, is around $0.5^\circ \pm 0.1667^\circ$ [9], which can use to prevent the noises that etalon effect cause [5], increase the correctness.

For the way of sealing for the gas cell, axial face sealing is using due to the mounting plate and the gas cell, so that two O-ring can push together with the compression force provide by the screw and the mounting plate, will makes the connection parts of the wedge window and the cell make vacuum [8], to isolate cell from the normal air.

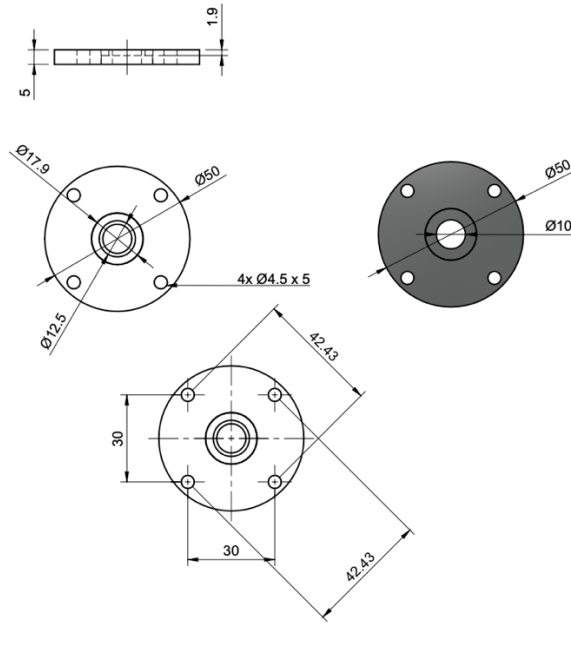
Approach to the problem

1. Select the IR laser source, as methane absorb specific wavelength of laser sources, so selecting the specific IR laser will be very important, as the research shows, the absorption line of methane wavelength range is from 1.65, 2.3, 3.33.4 and 7.7 μm [10], so while we doing the experiment, we need to use the IR laser with wavelength range from above.
2. Select the receiver for the IR laser, since the IR laser for the methane works with specific wavelength, so while we select the specific laser, choose appropriate receiver should also be one of the approaches to detect the intensity of the laser inside the gas cell by converting the transmittance to voltage.
3. Design and assemble the testing gas cell, the gas cell is asked to not exceed the length of 15 cm, so the design is based on ≤ 15 cm after assembling for the whole gas cell design, the whole gas cell includes 3 main parts. 1. The main gas cell with connected stand and male connector. 2. The 2 mounting plates with groove and 4 O-rings and wedge window. 3. The IR laser and Receiver.
4. Testing using the application, with the receiver receive the intensity of the laser inside the gas cell and present them using the data or plot graph.
5. Validate the system with checking the result with the Beer-Lamber Law or the SpectraPlot, and identify the cause of error may come out, debug if possible and see if any upgrade from the system or the programme can be made.

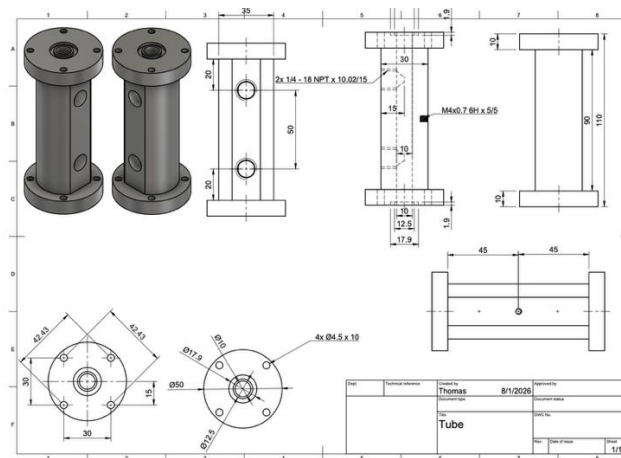
Results and discussions

Result:

1. Design of the methane gas cell.



(Figure 1)



(Figure 2)

1. The design of the gas cell
 1. The actual dimensions of the fabricated gas cell is same as the cad designed one, which is total 14.16cm length, and the width/diameter of the gas cell is 3cm, as we need to calculate the expected/theoretical absorption using the database with the main

body of the gas cell, such after getting the length of the main body, and that will be 11cm, which is the L for the Beer-Lambert Law.

2. There are 4 m4 holes for each side at the plate and the main gas cell, which is for fasten the plate, the reason of designing with this method is, in real engineering application, we may have many kinds of gases need to be testing, they may have different wavelength, such as carbon dioxide is $15\mu\text{m}$ and methane is $7.7\mu\text{m}$, while for those situation, we may need to adjust the laser and the optical equipment, such, the design I am choosing is for have more flexibility for adjusting to the system, for changing the wedge window and avoiding errors.
3. The reason of choosing the circular shape is mainly for 2 reasons. One is for experiment accuracy, the second is for budget control.
 1. For experiment accuracy, if we are choosing the rectangular shape for this gas cell, there may have some 90 degrees corner and flat surfaces, and when the IR laser pass through the cell, the surfaces may have some reflection, which generate the etalon effect [5] create error and noises, affect the experiment results.

Second, for the budget control, compare to different shape, most of the part before machining are in circular shape, so that if we choose to use other shape, we may need to use a larger part to work with it to become our desired shape, and this may increase the cost, and with using circular shape, we can save our time to do the fabrication and cost for making the shape, so choosing the circular shape is a better solution for the TDLAS experiment compare to the others.

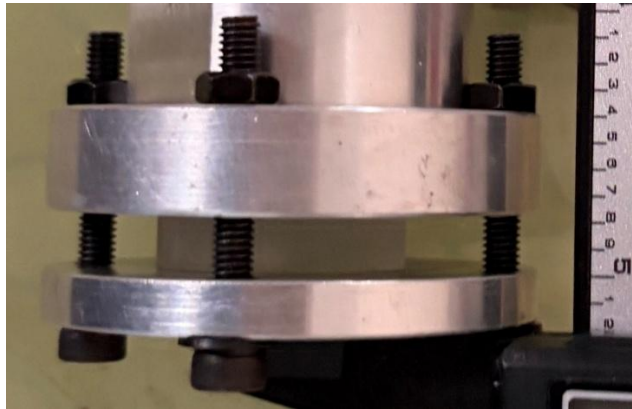
4. As for the surface of the NPT holes and the bottom m4 hole, the design is using flat surface milling, compare to the original circular shape surface, flat surface can reduce the error of positioning and ensure the same milling length of both of the hole, such that compare to the original circular shape surface using, using flat surface will have less error to affect the result.

2. Assemble of the gas cell

1. Before plugging the O-ring in to the slots, we need to fill some of the silicone between the them, the reason of it is because we need to improve the extrusion of the O-ring, to support the O-ring and don't let it squeeze out, and improve the sealing to the wedge window and affect the result we get.
2. Install the wedge window between the plates and the gas cell.

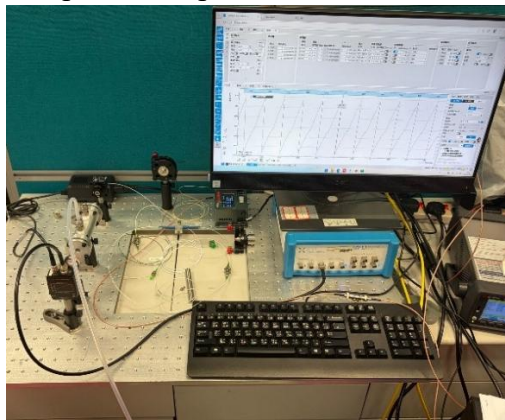
3. Using the screws fasten the plate and the gas cell for holding the wedge window
4. Cover the NPT male connectors with thread sealant to improve sealing and connect it to the above hole of the gas cell.

Such for one side of the gas cell setup after installment will become

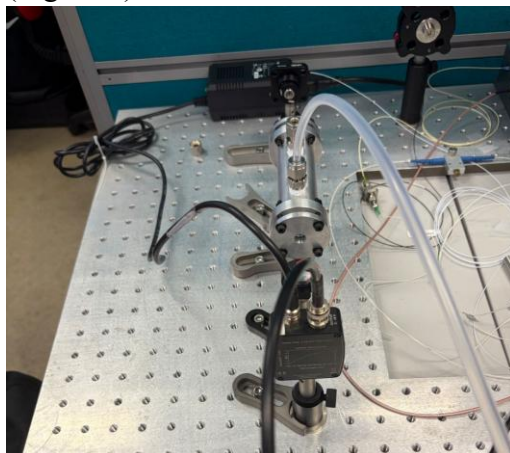


(Figure 3)

3. Setup of the experiment



(Figure 4)



(Figure 5)

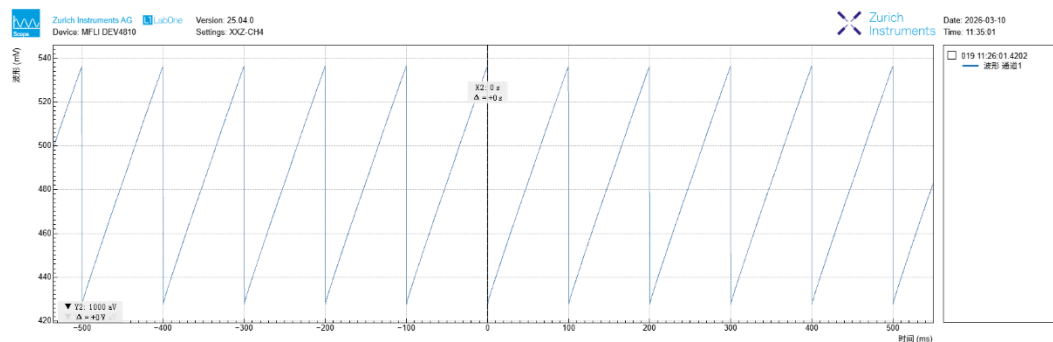
The experiment is using different concentration (I.e. 0, 2000, 4000, till 10000ppm) of methane to test how the voltage reacts when facing different concentration.

Measurements were conducted under standard ambient conditions (1atm, 300 K) to simulate a typical industrial environment and obtain representative data.

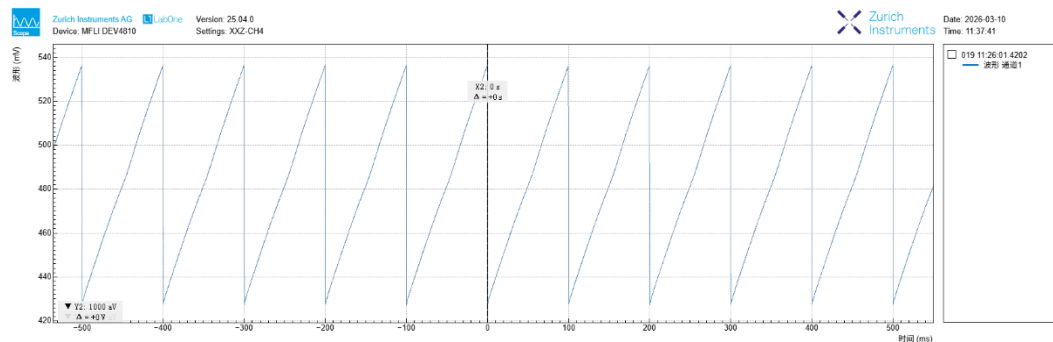
The schematic of the experiment is shown in the above figure, the gas cell plugged in with one side of the tube, which is use for transfer methane to the gas cell, and if there are more specific and high accuracy experiment is doing, the opposite side also have a NPT connector to let the tube to install a valve to make the gas cell become vacuum to do the test. For the test above by simulating the normal industrial environment use, which can have a more closer data similar to those situations.

As shown in the figure 4 at the front is the IR laser, the middle part is the gas cell, and at the back is the photodetector, the IR laser beam passes through the gas cell with containing the methane-air mixture before reaching the photodetector, and those absorbance are recorded with voltage signal, the raw waveform of different concentration are presented at Figure 5-10, after the waveform is recorded, the results will be extract out and transferred to out and become different plot using MATLAB.

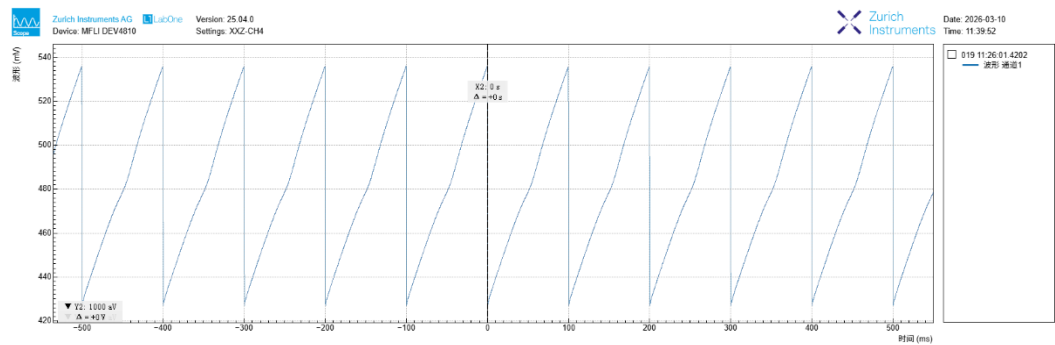
4. The result observed after experiment and testing.



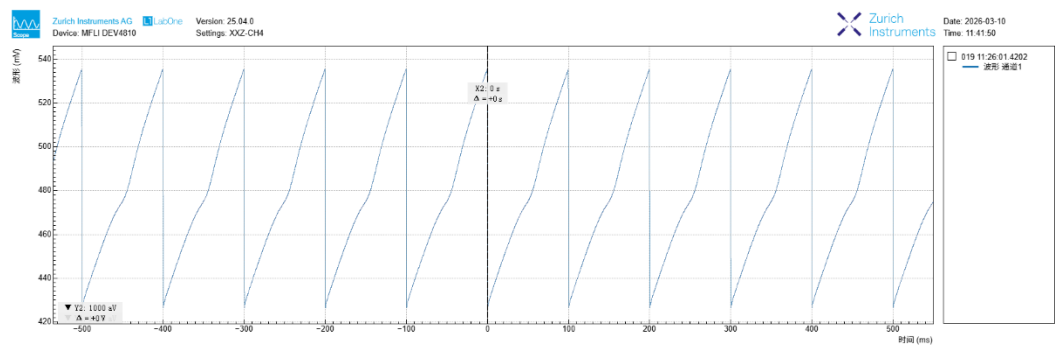
(Figure 6)



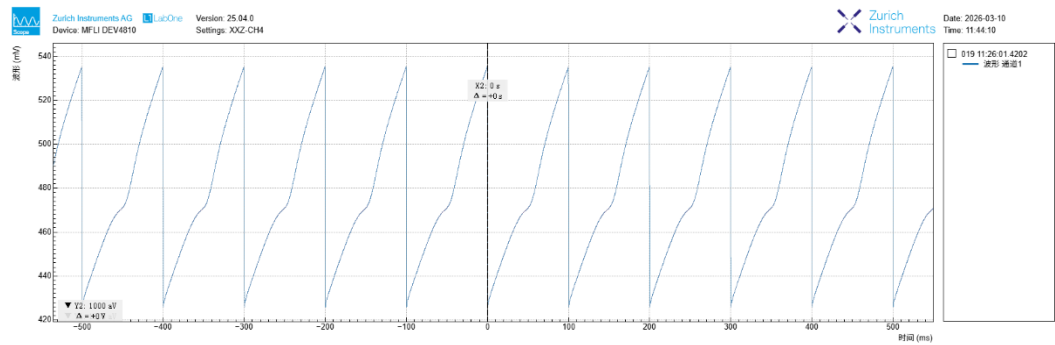
(Figure 7)



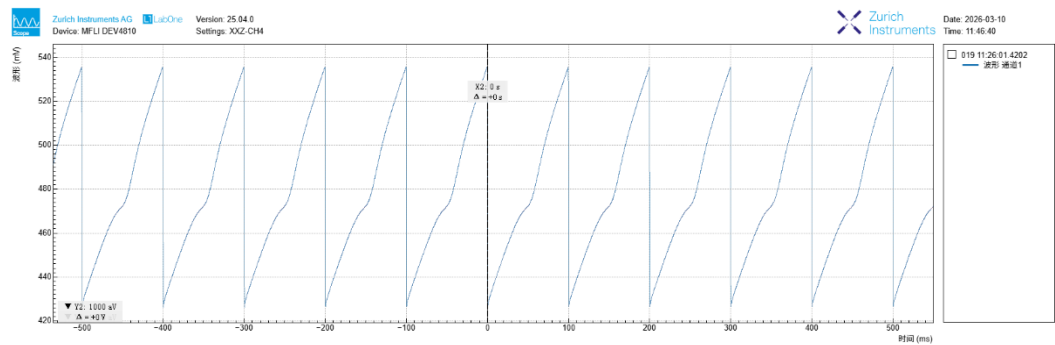
(Figure 8)



(Figure 9)



(Figure 10)



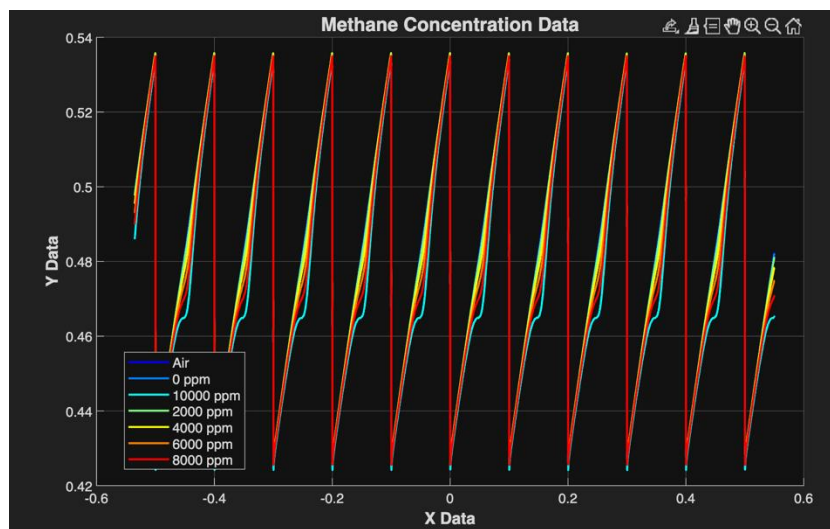
(Figure 11)

From the above figures, the originally shape of the curve is in linear form, and while the concentration rises, the linear shape start to pop up a bubble at the climbing stage within 100ms, and that bubble will shows the absorbance of the methane concentration rises.

Main phenomenon observed

1. At low concentration, (0-2000 ppm), the shape of the curve still keeping as clear linear form.
2. At mid concentration,(4000-8000 ppm), a clear shape of the bubble emerges, and affecting the linear shape of the original curve, while we use the Beer-Lamber Law or area difference to calculate, the absorbance and error is still reliable and acceptable.
3. At high concentration, (10000 ppm), the absorbance reach to the max, where the bubble is at max prominence, and the largest voltage drop, means that the strongest attenuation happens in the 10000ppm.

5. Combined results using MATLAB and AI.



(Figure 12)

The code is placed at the site [11]

By the result of figure 11, as the ppm in the gas cell increase, the area will also increase, and that area difference equals to the intensity of the gas in the cell, by getting this conclusion, we will need to know about how the intensity of the gas in the cell change will represent in the graph with calculating of the intensity.

There are two ways to calculate the absorbance of the gas in the cell.

1. By using Beer-Lambert Law [1]

$$\text{Absorbance } A = -\ln\left(\frac{I}{I_0}\right) = K_v L = \alpha C L$$

2. By calculating the area difference with integration.

$$\text{Area}_{air} = \int_{t_1}^{t_2} V_{air} dt, \text{Area}_{gas} = \int_{t_1}^{t_2} V_{gas} dt$$

Since we want area difference, by fundamental theorem of calculus

$$\text{Area}_{difference} = \int_{t_1}^{t_2} [V_{air}(t) - V_{sample}(t)] dt$$

And the reason for using both of the method to calculate is in some of the sense in the real application, different variations like temperature, pressure change will affect the stable of the baseline, so to increase the accuracy of the system, using both of the calculation and know them for different pros and cons, can increase the correctness.

By this idea we can ask AI to build base on this idea. With the code

1. Using Beer Law

```
% ===== Peak & Area Calculation =====
peak_range = [2190 2210];
idx_band = freq >= peak_range(1) & freq <= peak_range(2);

conc_ppm_full = conc_ppm(:);
A_peak = zeros(5,1);
A_area = zeros(5,1);

for k = 1:5
    A_peak(k) = max(A_all(idx_band, k));
    A_area(k) = trapz(freq(idx_band), A_all(idx_band, k));
end

% ===== FIGURE 2: Method 1 (Peak Absorbance) =====
figure(2);
set(gcf, 'Position', [500 100 600 500]);
plot(conc_ppm_full, A_peak, 'o', 'LineWidth', 2, 'MarkerSize', 8, 'Color', [0 0.447 0.741]);
hold on;
p_peak = polyfit(conc_ppm_full, A_peak, 1);
fit_y_peak = polyval(p_peak, conc_ppm_full);
plot(conc_ppm_full, fit_y_peak, 'r--', 'LineWidth', 2);

% R^2 Calculation
R2_peak = 1 - sum((A_peak - fit_y_peak).^2) / ((length(A_peak)-1) * var(A_peak));

xlabel('Concentration (ppm)');
ylabel('Peak Absorbance');
title('Method 1: Peak Absorbance Linearity');
legend('Experimental', sprintf('Fit R^2 = %.4f', R2_peak), 'Location', 'northwest');
grid on;
```

(Figure 14) [12]

2. For area difference


```

% ===== FIGURE 3: Method 2 (Integrated Area) =====
figure(3);
set(gcf, 'Position', [900 100 600 500]);
plot(conc_ppm_full, A_area, 's', 'LineWidth', 2, 'MarkerSize', 8, 'Color', [0.85 0.325 0.098]);
hold on;
p_area = polyfit(conc_ppm_full, A_area, 1);
fit_y_area = polyval(p_area, conc_ppm_full);
plot(conc_ppm_full, fit_y_area, 'b--', 'LineWidth', 2);

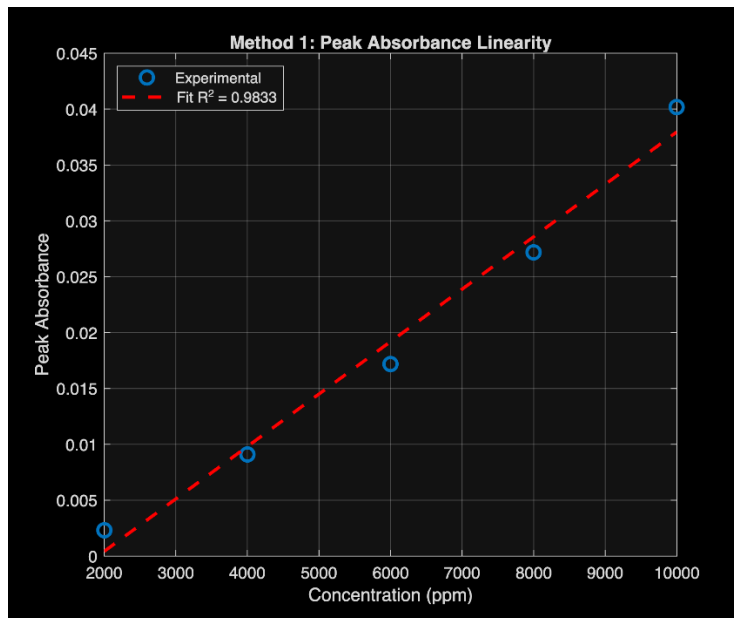
% R^2 Calculation
R2_area = 1 - sum((A_area - fit_y_area).^2) / ((length(A_area)-1) * var(A_area));

xlabel('Concentration (ppm)');
ylabel('Integrated Area (cm^-1)');
title('Method 2: Integrated Area Linearity');
legend('Experimental', sprintf('Fit R^2 = %.4f', R2_area), 'Location', 'northwest');
grid on;

```

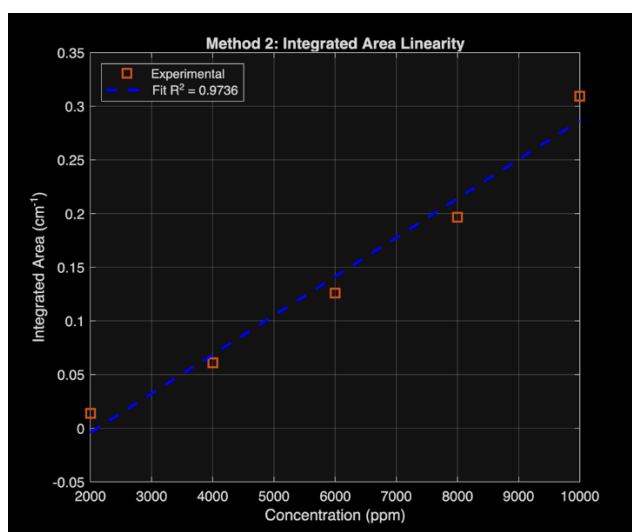
(Figure 13) [12]

After using the code to extract the results and calculate the absorbance, which come out results of that, and plotting it using the graph.



(Figure 15)[12]

Concentration	Absorbance
0ppm	0
2000ppm	0.00230877
4000ppm	0.00908884
6000ppm	0.0171923
8000ppm	0.0271742
10000ppm	0.040186
R^2	0.9833



(Figure 16) [12]

Concentration	Absorbance
0ppm	0
2000ppm	0.0137614
4000ppm	0.0609371
6000ppm	0.125966
8000ppm	0.196626
10000ppm	0.309409
R ²	0.9736

These 2 result shows that in 2000-8000 ppm section, the absorbance of methane in the gas cell will work as a basically linear relationship, as both of the figure shows the R^2 is around 0.9833 and 0.9736, which is around 0.979 , shows up the curve is quite linear, and can be accept as a linear relation, but there may still have some of the noises or cause or error may happen lead to this result.

Also, while the concentration rises, the absorbance will rises at the same time, but while 10000 ppm the linearity doesn't show up very visibly at both of the calculations with different methods, there may have two reasons may lead to this result.

First, to avoid the misuse/miscalculation with the formula, according to Beer Law, there is one important prerequisite for using the law, which is

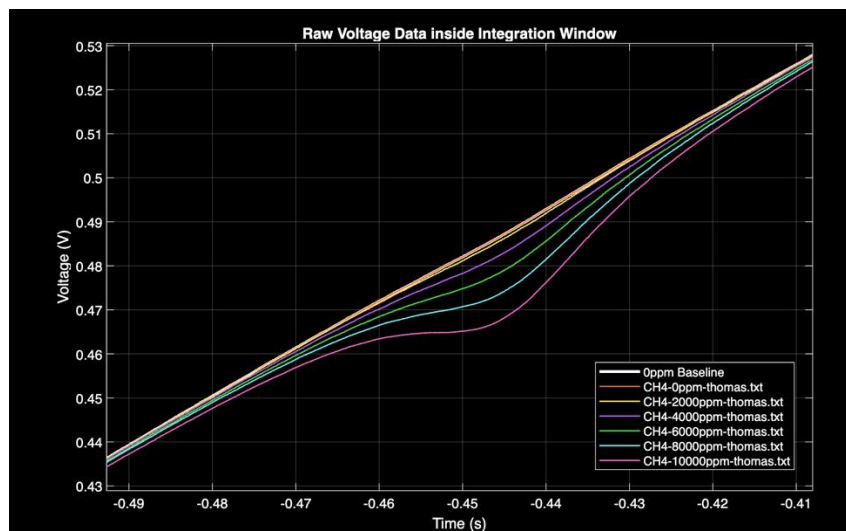
1. The absorbance A should be smaller than 1, to make that is linearity.

As by the code we already got $A = 0.040186$ or 0.309409 for 10000 ppm, which both calculation method is eligible for the condition which $A < 1$, so, to explain this phenomenon, there may appear pressure broadening or Baseline

Fitting Failure happen at the curve we got while at the 10000-ppm concentration stage.

1. For pressure broadening, as the concentration rises, the mole amount of methane in the gas cell increases, and the collision will become more frequent, makes the pressure broadening happened, according to Razavipour et al [13]. The pressure broadening happened will increase the line width at the absorption line and decrease the peak absorbance, as the phenomenon is also show at the Figure 10, compare to the low concentration, the two side of the bubble is more extend and the increase of the peak is slower than the low concentration, and the pressure broadening will affecting our calculation, and make it become cause of error in the plotting.
2. For Baseline Fitting Failure, by the impact for pressure broadening, as we are using the baseline calculate the area difference for method 2, but after the pressure broadening, the baseline has been offset, and by the peak absorbance has become smaller, the calculation may have mistaken by using both method above, which makes the absorbance offset from the linearity relationship, and this result is similar to the research by Weisberger et al [14], they state that the traditional baseline fitting method will have error on high concentration due to without enough no absorption zone, which leads to the error due to the baseline fitting failure.

As to more explain these two, by asking AI to zoom in to compare the one absorption zone of each of the concentration.



(Figure 17) [15]

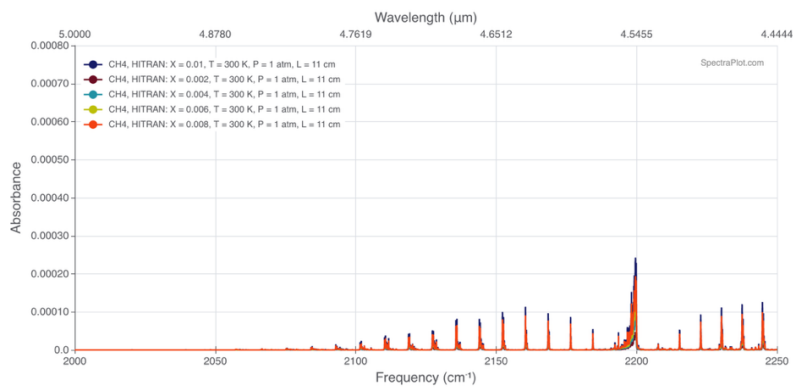
From the figure, the baseline in the low concentration can still be accept to calculate the absorbance, but while the concentration increase and accumulate the error to the maximum at 10000ppm, and also it is the

most offset concentration, so by this situation, the pressure broadening will lead to baseline failure and affect our calculation using traditional method.

5. Standard absorption setup of methane with database using Spectraplot

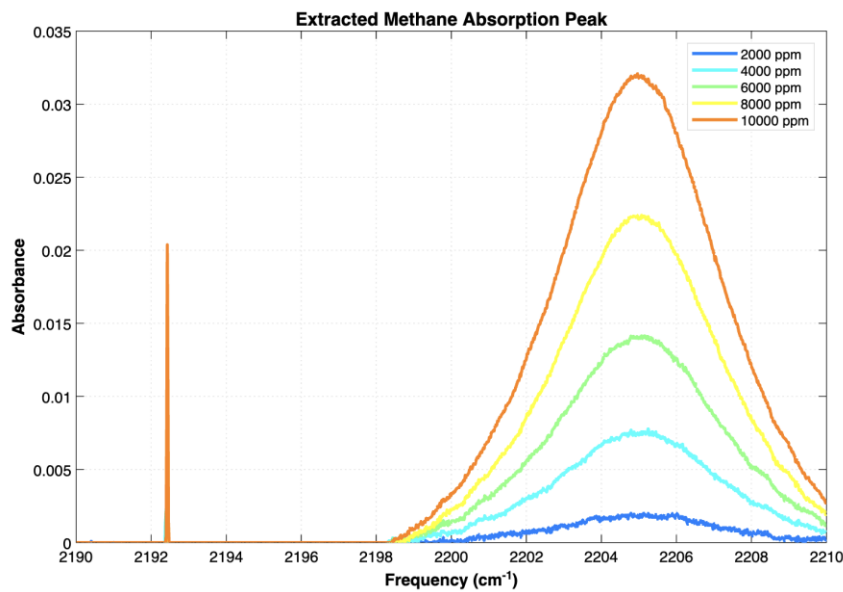
Species	T (K)	P (atm)
Methane	300	1
Databases	X_{CH_4}	L (cm)
HITRAN	0.01	11

(Figure 18) [16]



(Figure 19)[16]

By asking AI to extract the absorbance spectrum and use the databased of HITRAN from SpectraPlot to compare.



(Figure 20) [18]

The discussion of figure can compare to the result get using MATLAB, which is figure 13, at both figure around 2200 cm^{-1} the absorbance from different concentration will have a wave and reach to the peak, shows that the main absorbance characteristic position remains similar to both figures.

For the similarities, both figure at 10000 ppm, will start to have the rising trend of the absorbance from 2190 cm^{-1} , and while for our experiment, the max absorbance got from the experiment is around 0.0325 and > 0.00030 from the databases.

The key difference, from the Figure of the HITRAN database, there have some small signal starting to appear significantly from 2100 cm^{-1} , and for the experiment one which only appear once at 2193 cm^{-1} , shows that the experiment more focus on the main wave, and hard to show other small signals appear in the experiment, which makes this as the main difference of the database and the observation.



(Figure 21)

The leakage of the system can be reference with the vacuum gauge, and due to time limitation of the experiment, 10 minutes of testing is tested with the system.

Minute	Torr
0	4
2	12
4	16
6	21
8	26
10	31

Improvements and errors

Improvements

1. The pitch of the hole of the NPT male connector, since this time is using the NPT thread hole, and lack of experience of designing the special type of threads, such that which makes while the connector plugged in the hole, they cannot be fully twisted inside the hole, only half of them are plugged, although in the normal pressure environment this type of situation can be accepted, but while in the in the vacuum situation at the leakage test, the system is at 31 torr after 10 minutes, compare to the tutor's demo is around single digit, which shows that system may have a leakage of the gas due to this reason, so the experiment equipment can be further optimize, such as referencing online to see how the others design a special thread hole, so that for problem at the pitch, may lead to errors and different results and affect the accuracy.
2. To have a more improvement on reducing the etalon effect happens in the experiment, since the absorbance is very little after getting the result, such that to have a more accurate result of experiment, the way of improve this can be preparing different set of equipment, to check if the data are similar, which can increase the accuracy of the results.
3. Since testing in the above experiment is in ambient environment, and within the normal air, there some of the methane exist, the movement the methane in air may cause the noises the appear in the data, and affect the result, such for improving this situation, we may use the vacuum situation inside the gas cell and inserting the methane gas to check the data to reduce the noise as much as possible.
4. Since we know there is pressure broadening and baseline fitting failure may happen in our experiment, the improvement of this can be we can adjust our program, from the reference of the BFBL method from Weisberger et al. (2018) [14] they suggest that the step of BFBL method is.
 1. Using the HITRAN database and correct linear type and calculate the simulated transmittance spectrum.
 2. Then using the transmittance to calculate the baseline anchor voltage and fitting the voltage to a no absorption baseline.
 3. Then use the baseline to calculate the actual transmittance and continue to adjust, which can increase the correctness of the baseline fitting.

Extra discussing question

1. If we are putting the system in the real-life industry environment, there may have some of the vibration and cause the connection of the screw is loose, and reduce the sealing of the wedge, what can we do to reduce this situation.

Ans: For reducing the loose of the screw, we can add some anti vibration washer inside the screw, the special of the washer is, they will have some small protrusions and increase the friction of screw while twisting, so that which can make the screw harder to loose and stabilize correctness of system.

2. In real life industrial application, while the engineering work has start, and the system is brought to do the monitoring for the gas, what is the effect that temperature change brings to the result, and what can we do to reduce the effect as much as possible.

Ans:

First, the temperature will affect the line strength, leading to different absorbance, according to HITRAN database [17] and use the Beer Law in the textbook [1].

$$A(\nu) = S(T) \times \phi(\nu) \times C \times L$$

And $S(T)$ according to HITRAN [17] and Hanson et al [1], can be express as

$$S_i(T) = S_i(T_0) * (Q(T_0)/Q(T)) * (T_0/T) * \exp[-(hcE_i''/k) * (1/T - 1/T_0)] \\ * [1 - \exp(-h\nu_{0,i}/kT)] / [1 - \exp(-h\nu_{0,i}/kT_0)]$$

Such that if T increase

1. $Q(T)$ increase and lead to $(Q(T_0)/Q(T))$ decrease
2. $\exp[-(hcE_i''/k) * (1/T - 1/T_0)]$ will be decrease

Then the Absorbance A will be reduced and affected, leading to a wrong curve relationship, for improving this situation, we can add a temperature control to the gas cell, which can manage the temperature inside the cell, and limit the variance of the temperature to reduce the error.

Conclusion

To conclude, the formula of Beer-Lambert Law shows can be appear in the result observed in this TDLAS experiment testing in the ambient environment, the linear relation R^2 observed is around 0.979, which is calculated by using the Beer-Lambert Law method and the area difference method to calculated the absorbance.

The result of $R^2 = 0.979$ shows that, the result absorbance curve is shown as a basically linear relationship, and can be accepted, though there may have some noises from the environment and the other cause of error, but still, it is able to proof the correctness of Beer-Lambert Law.

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